

The bridge midpoint on a compact group at any cut: windings and the centre

Result, 2026-09-28 (Claude; written derivation, to be refereed by GPT-6 Astra).

Let G be a compact simply connected Lie group and take the heat-kernel bridge of total heat time t from e to $g = e^{iX}$, cut at fraction $s \in (0, 1)$, so that the first piece has heat time st . For every irreducible representation λ the midpoint expectation of its character is an exact finite sum, over the weights κ of λ , of image sums over the coroot lattice $Q^\vee$ (Theorem 1):

$$E \chi_\lambda(m) = \frac{\sum_{\kappa} m_\lambda(\kappa) e^{-ts(1-s)|\kappa|^2/2} e^{is\langle\kappa, X\rangle} \sum_{H \in Q^\vee} e^{2\pi i(1-s)\langle\kappa, H\rangle} \pi\left(\frac{i}{t}(X - 2\pi H) - (1-s)\kappa\right) G_H}{\sum_{H \in Q^\vee} \pi\left(\frac{i}{t}(X - 2\pi H)\right) G_H}, \quad G_H = e^{-|X-H|^2/4t} \quad (1)$$

with $m_\lambda(\kappa)$ the weight multiplicities and $\pi(v) = \prod_{\alpha > 0} \langle v, \alpha \rangle / \langle \rho, \alpha \rangle$ the Weyl dimension polynomial.

- **Corollary 2 (reductions).** For $U(1)$, (1) is the winding mixture of Corollary 5_s(a). For $SU(2)$ at $s = \frac{1}{2}$ it is Theorems 1 and 3 of the $SU(2)$ midpoint note, image terms included.
- **Corollary 3 (the centre).** The phase $e^{2\pi i(1-s)\langle\kappa, H\rangle}$ of an image is the same for all weights of every representation iff $(1-s)H \in P^\vee$, the coweight lattice, and it is then the central character of λ at $z_H = e^{2\pi i(1-s)H}$. At $s = p/q$ in lowest terms these central images reach exactly the q -torsion of the centre: $\mathbb{Z}_{\gcd(q, N)}$ for $SU(N)$.
- **Corollary 4 ($SU(3)$).** At $s = \frac{1}{3}$ the central images are $H \in 3P^\vee$; the smallest are the two Weyl triples of $\pm(1, 1, -2)$, of relative weight about $e^{-24\pi^2/t}$, and they carry ω^2 and ω on the fundamental ($\omega = e^{2\pi i/3}$), the conjugates on the antifundamental and 1 on the adjoint. At $s = \frac{1}{2}$ every image with a nontrivial phase has a weight-dependent one: no halving of $SU(3)$ reaches the centre.

For $SU(2)$ at $s = \frac{1}{2}$ the phase is $(-1)^w$ for half-integer spin and 1 for integer spin. This is Dirac's belt trick, $\pi_1(SO(3)) = \mathbb{Z}_2$ (Newman 1942, metadata), appearing in the exact midpoint law: a bridge winding w times round the group makes w extra 720° turns, and its midpoint makes w extra 360° turns, which the spinor characters detect. For $SU(3)$ the same mechanism carries the triality, and only cuts at thirds (or at denominators divisible by 3) produce it.

The ingredients are character orthogonality, the Brauer–Klimyk rule and Poisson summation, as in the $SU(2)$ note; the denominator of (1) is the classical image formula for the heat kernel on a compact group (Fegan 1983, metadata). No novelty is claimed for the method.

1. Conventions

The group metric is that of the series/parallel note §1: $-\Delta_G \chi_\lambda = C_2(\lambda) \chi_\lambda$ with $C_2(\lambda) = |\lambda + \rho|^2 - |\rho|^2$ in the inner product on weights dual to the metric on the Cartan algebra $\mathfrak{t}$. For $SU(N)$ roots have $|\alpha|^2 = 1$ and coroots $\alpha^\vee = 2\alpha/|\alpha|^2$ have $|\alpha^\vee|^2 = 4$, which is the metric $|Y|^2 = 2 \operatorname{tr} Y^2$ on $\mathfrak{t}$; then $C_2(j) = j(j+1)$ for $SU(2)$ and $C_2(\mathbf{3}) = \frac{4}{3}$ for $SU(3)$. Torus elements are e^{iX} , $X \in \mathfrak{t}$; the pairing of the weight lattice P with the coroot lattice $Q^\vee$ is integral, P and $Q^\vee$ are dual lattices, and $e^{iX} = e$ iff $X \in 2\pi Q^\vee$ (G simply connected). The coweight lattice $P^\vee$ is dual to the root lattice Q , and $P^\vee/Q^\vee \cong Z(G)$ through $Y \mapsto e^{2\pi i Y}$. For $SU(2)$, $|X| = \theta$ is the rotation angle of the $SU(2)$ note, and $X - 2\pi w \alpha^\vee$ has length $|\theta - 4\pi w|$.

Heat kernel $k_t = \sum_{\lambda} d_{\lambda} e^{-tC_2(\lambda)/2} \chi_{\lambda}$. The bridge midpoint m has density $k_{st}(m) k_{(1-s)t}(m^{-1}g) / k_t(g)$ with respect to Haar measure; the bridge law (4) of the series/parallel note becomes this one after a left translation, and Corollary 2_s there gives the cut faces heat times st and $(1-s)t$.

Weyl: $A_v(X) = \sum_{w \in W} \varepsilon(w) e^{i\langle wv, X \rangle}$, $\chi_\lambda = A_{\lambda+\rho}/A_\rho$, $d_\lambda = \pi(\lambda + \rho)$. Extend $\chi_\nu := A_{\nu+\rho}/A_\rho$ to every $\nu \in P$: it is zero or $\pm$ an irreducible character, and $C_2(\nu) := |\nu + \rho|^2 - |\rho|^2$ is invariant under the shifted Weyl action, so it is the Casimir of that character.

Brauer–Klimyk rule. $\chi_\lambda \chi_\mu = \sum_\kappa m_\lambda(\kappa) \chi_{\mu+\kappa}$ for dominant μ . Indeed $\chi_\lambda A_{\mu+\rho} = \sum_\kappa m_\lambda(\kappa) e^{i\langle \kappa, X \rangle} \sum_w \varepsilon(w) e^{i\langle w(\mu+\rho), X \rangle}$, and replacing κ by $w\kappa$, which leaves m_λ invariant, gives $\sum_\kappa m_\lambda(\kappa) A_{\mu+\kappa+\rho}$.

2. Theorem 1 and its proof

Theorem 1. For X regular, (1) holds; for singular X it holds by continuity.

Proof. (i) *Characters.* Expanding both heat kernels, $k_t(g) E \chi_\lambda(m) = \sum_{\mu, \nu} d_\mu d_\nu e^{-stC_2(\mu)/2 - (1-s)tC_2(\nu)/2} \int \chi_\lambda$. The Brauer–Klimyk rule and $\int \chi_a(m) \chi_b(m^{-1}g) dm = \delta_{ab} \chi_a(g)/d_a$ give

$$k_t(g) E \chi_\lambda(m) = \sum_{\mu \text{ dominant}} d_\mu \sum_{\kappa} m_\lambda(\kappa) e^{-stC_2(\mu)/2 - (1-s)tC_2(\mu+\kappa)/2} \chi_{\mu+\kappa}(g).$$

(ii) *Unfolding.* Multiply by $A_\rho(X)$ and put $v = \mu + \rho$, which runs over the strictly dominant weights. The summand becomes $\pi(v) m_\lambda(\kappa) e^{-\frac{t}{2}[s|v|^2 + (1-s)|v+\kappa|^2 - |\rho|^2]} A_{v+\kappa}(X)$. Expand $A_{v+\kappa}$ and substitute $v \mapsto w^{-1}v$, $\kappa \mapsto w^{-1}\kappa$: $\varepsilon(w)\pi(w^{-1}v) = \pi(v)$, and m_λ and the norms are Weyl invariant. Each regular weight is the image of exactly one strictly dominant weight, and π vanishes on the walls, so

$$A_\rho(X) k_t(g) E \chi_\lambda(m) = e^{t|\rho|^2/2} \sum_{\kappa} m_\lambda(\kappa) \sum_{v \in P} \pi(v) e^{-\frac{t}{2}[s|v|^2 + (1-s)|v+\kappa|^2]} e^{i\langle v+\kappa, X \rangle}.$$

(iii) *Square.* $s|v|^2 + (1-s)|v+\kappa|^2 = |v + (1-s)\kappa|^2 + s(1-s)|\kappa|^2$.

(iv) *Poisson summation* over P , whose dual lattice is $Q^\vee$: $\sum_{v \in P} f(v) = \text{vol}(\mathfrak{t}^*/P)^{-1} \sum_{H \in Q^\vee} \hat{f}(H)$ with $\hat{f}(H) = \int f(y) e^{-2\pi i \langle y, H \rangle} dy$. With $y = u - (1-s)\kappa$ one has $\langle y+\kappa, X \rangle = \langle u, X \rangle + s\langle \kappa, X \rangle$ and $\langle y, H \rangle = \langle u, H \rangle - (1-s)\langle \kappa, H \rangle$, so

$$\hat{f}(H) = e^{is\langle \kappa, X \rangle} e^{2\pi i(1-s)\langle \kappa, H \rangle} \int \pi(u - (1-s)\kappa) e^{-t|u|^2/2} e^{i\langle u, Z \rangle} du, \quad Z = X - 2\pi H.$$

(v) *The Gaussian integral.* π is harmonic: $\Delta \pi$ is Weyl-antisymmetric of lower degree, and every Weyl-antisymmetric polynomial is divisible by π . Its derivatives are harmonic too. For a harmonic homogeneous h of degree d , Hobson's formula $h(\partial)F(|Z|^2) = 2^d h(Z) F^{(d)}(|Z|^2)$ gives $h(-i\partial_Z) e^{-|Z|^2/(2t)} = h(iZ/t) e^{-|Z|^2/(2t)}$. Taylor-expanding $\pi(u-c)$ in c into harmonic pieces and summing back,

$$\int \pi(u-c) e^{-t|u|^2/2 + i\langle u, Z \rangle} du = \left(\frac{2\pi}{t}\right)^{n/2} \pi\left(\frac{iZ}{t} - c\right) e^{-|Z|^2/(2t)}, \quad n = \dim \mathfrak{t}.$$

(vi) The trivial representation ($\kappa = 0$ only) gives the denominator. The constants $e^{t|\rho|^2/2}$, $\text{vol}(\mathfrak{t}^*/P)^{-1}$ and $(2\pi/t)^{n/2}$ cancel. All sums converge absolutely. $\square$

The polynomial ratio for a single image is $\pi(\frac{i}{t}Z - (1-s)\kappa)/\pi(\frac{i}{t}Z) = \prod_{\alpha > 0} (1 + i(1-s)t\langle \kappa, \alpha \rangle / \langle Z, \alpha \rangle)$, the curvature correction; for $SU(2)$ at $s = \frac{1}{2}$ it is the factor $1 + ikt/(2\theta)$ behind the $-\frac{t}{2\theta}k \sin \frac{k\theta}{2}$ term of the $SU(2)$ note.

3. Reductions (Corollary 2)

$U(1)$. $P = Q^\vee = \mathbb{Z}$, $\pi \equiv 1$, $\rho = 0$, $\langle n, H \rangle = nH$, $X = \theta$: $E e^{inm} = e^{-ts(1-s)n^2/2} \sum_H e^{in(s\theta+2\pi(1-s)H)} G_H / \sum_H G_H$ the characteristic function of the mixture over the winding H , with weights $\propto e^{-(\theta-2\pi H)^2/(2t)}$, of wrapped Gaussians of variance $s(1-s)t$ centred at $s\theta + 2\pi(1-s)H$. This is Corollary 5_s(a) of the series/parallel note.

$SU(2)$, $s = \frac{1}{2}$. The weights of spin J are $\kappa = k\alpha$, $k = -J, \dots, J$, with $|\kappa|^2 = k^2$, $\langle \kappa, X \rangle = k\theta$, $\langle \kappa, w\alpha^\vee \rangle = 2kw$, and $\pi(v) = 2\langle v, \alpha \rangle$. The phase is $e^{2\pi i k w}$: $(-1)^w$ for half-integer J and 1 for integer J . With $\Xi_{(\sigma)}, \Theta_{(\sigma)}$ as in the $SU(2)$ note, the numerator of (1) is $\sum_k e^{-tk^2/8} e^{ik\theta/2} [\frac{i}{t}\Xi_{(\sigma)} - \frac{k}{2}\Theta_{(\sigma)}]$ and the denominator is $\frac{i}{t}\Xi_+$. Symmetrizing in k gives Theorem 3 of that note, and $J = \frac{1}{2}$ gives its Theorem 1, $e^{-t/32} [2 \cos \frac{\theta}{4} \Xi_- - \frac{t}{2} \sin \frac{\theta}{4} \Theta_-] / \Xi_+$.

4. The centre (Corollary 3)

Proof. Two weights of one representation differ by an element of the root lattice Q , so $e^{2\pi i(1-s)\langle \kappa, H \rangle}$ is independent of κ for every λ iff $\langle \beta, (1-s)H \rangle \in \mathbb{Z}$ for all $\beta \in Q$, i.e. iff $(1-s)H \in Q^* = P^\vee$. Then $e^{2\pi i(1-s)\langle \kappa, H \rangle} = e^{2\pi i\langle \lambda, (1-s)H \rangle}$ is the scalar by which the central element $z_H = e^{2\pi i(1-s)H}$ acts in λ . For $s = p/q$ in lowest terms, $(1-s) = (q-p)/q$ with $\gcd(q-p, q) = 1$, and $(q-p)H/q \in P^\vee$ iff $H/q \in P^\vee$ (Bézout, since $H \in Q^\vee \subset P^\vee$). So the central images are $H \in qP^\vee \cap Q^\vee$, and the elements reached are $e^{2\pi i(q-p)Y}$ with $Y \in P^\vee \cap q^{-1}Q^\vee$: the q -torsion of $P^\vee/Q^\vee \cong Z(G)$, on which multiplication by $q-p$ is an automorphism. For $SU(N)$, $Z(G) = \mathbb{Z}_N$ and its q -torsion is $\mathbb{Z}_{\gcd(q, N)}$. $\square$

Consequences. Dyadic refinement ($q = 2^n$) reaches the 2-primary part of the centre: all of it for $SU(2)$ at the first halving, ± 1 at a halving and $\mathbb{Z}_4$ at quarters for $SU(4)$, nothing for $SU(3)$ or any odd N . The image classes of a halving are all central only for $SU(2)$, where $\frac{1}{2}Q^\vee = P^\vee$.

Geometrically the central elements are focal points of e : the shortest geodesics from e to $e^{2\pi i Y}$ ($Y \in P^\vee$ of minimal norm in its class) form the adjoint orbit of Y , a sphere S^2 for $-1 \in SU(2)$ and a $\mathbb{CP}^2$ for $\omega \cdot 1 \in SU(3)$, because conjugation fixes both endpoints.

5. $SU(3)$ (Corollary 4)

Use $\mathfrak{t} = \{H \in \mathbb{R}^3 : \sum H_i = 0\}$, $Q^\vee = \mathfrak{t} \cap \mathbb{Z}^3$, $|H|^2 = 2 \sum H_i^2$. The weights of $\mathbf{3}$ are $\kappa_i = e_i - \frac{1}{3}(1, 1, 1)$, so $\langle \kappa_i, H \rangle = H_i$: the image H carries the phase $e^{2\pi i(1-s)H_i}$ on the i -th weight.

- $s = \frac{1}{3}$: the phases $e^{4\pi i H_i/3}$ agree iff $H_1 \equiv H_2 \equiv H_3 \pmod{3}$, i.e. $H \in 3P^\vee$. For $(1, 1, -2)$ and its Weyl images all $H_i \equiv 1$ and the phase is $e^{4\pi i/3} = \omega^2$; for $(-1, -1, 2)$ and its images it is ω . The antifundamental gets the conjugates; the adjoint, with weights $e_i - e_j$, gets $e^{4\pi i(H_i - H_j)/3} = 1$. These images have $|H|^2 = 12$, so $G_H/G_0 = e^{-24\pi^2/t + 2\pi\langle X, H \rangle/t}$. The root images, such as $(1, -1, 0)$ with $|H|^2 = 4$ and $G_H/G_0 \approx e^{-8\pi^2/t}$, give the fundamental the phases $\omega^2, \omega, 1$: weight-dependent.
- $s = \frac{1}{2}$: the phases $e^{\pi i H_i}$ agree iff all H_i have the same parity; with $\sum H_i = 0$ they are then all even, $H \in 2Q^\vee$, and the phase is 1.

So an $SU(3)$ halving produces only weight-dependent phases (non-central shifts), and a trisection produces the triality character on the classes $3P^\vee$, below the root images by the factor $e^{-16\pi^2/t}$.

6. What the result says, and what it leaves open

It is an exact statement about one bridge, the building block of the parallel insertion, for every compact simply connected group, every representation and every cut fraction. It concerns characters only: the diagonal entries of the midpoint matrix $E D^\lambda(m)$ differ from the weight terms of (1), since Theorem 4 of the $SU(2)$ note gives the spin-1 entries an additional Gaussian

tail, so atlas cell 1 is unaffected. In the refinement it fixes which windings of a single step act through the centre. The weights are small per step: in $1 + 3$ the image exponents $8\pi^2$ (roots) and $24\pi^2$ (central $SU(3)$ trisection images) lie far beyond the per-volume threshold $c > 2/b_0 = 32\pi^2/11$ of the zero-spacing note for $SU(3)$. Centre-valued link configurations exist on any lattice whatever the blocking, and thick centre vortices (Mack and Petkova 1979; 't Hooft 1978; metadata) are a question about many steps. Whether a triadic refinement, $b = 3$, organizes the $SU(3)$ large-field terms better than dyadic refinement is open; factor-2 decimation with the $\mathbb{Z}_2$ factor kept explicit goes back to Tomboulis (1981) (metadata), and prior art on $b = 3$ or centre-adapted blocking for $SU(3)$ has not been searched.

7. Consequence for STATE

Atlas §1b: the bullet on which part of the centre a cut reaches now rests on Corollary 3 of an exact formula, (1), which also carries the $SU(2)$ belt trick and the $SU(3)$ triality at thirds. Refereeing by Astra is pending; the exact trisection formula for the fundamental of $SU(3)$ is (1) with $s = \frac{1}{3}$ and the weights of §5.